THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · MEDIUM LEVEL UNIT TEST SET 3

Chapters 1–5: **Patterns in Mathematics, Lines and Angles, Number Play, Data Handling and Presentation, Prime Time** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Powers of 2: $2, 4, 8, 16, \ldots$ The 7th term is: 1

(a) 126 **(b)** 130 **(c)** 128 **(d)** 192

2. Which angle type is the smaller angle between the hands of a clock at 3:00? 1

(a) straight **(b)** acute **(c)** obtuse **(d)** right

3. A Kaprekar step on 6174 gives: 1

(a) 6183 **(b)** 6165 **(c)** 6174 **(d)** 6264

4. On a bar graph with scale 1 cm = 5 units a bar is 6 cm. If its value increases by 20 units, the new height (cm) is: 1

(a) 11 **(b)** 9 **(c)** 10 **(d)** 12

5. The sum of all prime factors (distinct) of 210 is: 1

(a) 16 **(b)** 18 **(c)** 17 **(d)** 19

6. The nth term of the pattern $9, 15, 21, 27, \ldots$ is $a + (n-1)d$. The 15th term is: 1

(a) 93 **(b)** 94 **(c)** 99 **(d)** 87

7. Two angles of 40° and 65° are placed adjacent to each other. The angle left to complete a straight angle is: 1

(a) 75 **(b)** 85 **(c)** 80 **(d)** 65

8. The greatest 4-digit number whose digit sum is 10 is: 1

(a) 9190 **(b)** 9100 **(c)** 9091 **(d)** 9109

9. The total of the two largest values in the table is: 1

Item	Number
Orange	60
Grapes	40
Guava	10
Apple	40
Banana	45

(a) 110 **(b)** 105 **(c)** 115 **(d)** 100

10. The number of factors of 54 is: 1

(a) 7 **(b)** 9 **(c)** 8 **(d)** 10

11. In the pattern $7, 11, 15, 19, \ldots$ which term number equals 103? 1

(a) 27 **(b)** 25 **(c)** 24 **(d)** 26

12. What is the smaller angle between the hands of a clock at 4:20? 1

(a) 20 **(b)** 5 **(c)** 10 **(d)** 15

13. The smallest 4-digit number with digits 5, 6, 7, 2 (each used once) that is EVEN is: 1

(a) 2567 **(b)** 2576 **(c)** 2666 **(d)** 2585

14. A pictograph has one symbol = 25. The number of symbols for A is 8 and for B is 3. How many more does A have than B? 1

(a) 125 **(b)** 150 **(c)** 100 **(d)** 126

15. The HCF of 45 and 75 is: 1

(a) 14 **(b)** 15 **(c)** 16 **(d)** 17

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Find the sum of the first 30 odd numbers and the first 30 even numbers. 2

17. How much does the hour hand turn from 1:00 to 5:00? How much does the minute hand turn in that time? 2

18. Estimate 8764 - 3198 by rounding to the nearest hundred and find the actual difference. 2

19. Prepare a frequency table for the data 6, 5, 1, 3, 6, 6, 1, 2, 5, 6, 5, 2, 6, 3, 3, 6, 3, 6 and say which value is repeated the most. 2

20. Find the missing digit so that 456_ is divisible by 9. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. What is the greatest number that divides 48 and 72 exactly? How many equal groups of each kind can be made? 3

22. How many triangles' worth of sticks: a row of 15 triangles needs 3 sticks for the first triangle and 2 more for each new one. Find the sticks needed and the number of triangles that can be made with 41 sticks. 3

23. At what angle are the hands of a clock inclined at 3:20? Is the angle acute, obtuse or reflex (looking at the larger angle too)? 3

24. Perform two Kaprekar steps on 2863. 3

OR

Find the fraction of students liking Tennis and the number who do not like it. 3

Sport	Students
Tennis	20
Cricket	25
Hockey	35
Kabaddi	30

25. Find the prime factorisation of 200 and the number of its factors. 3

OR

Find the sum of all odd numbers from 1 to 79. Hence find the sum of all even numbers from 2 to 80. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. A school survey data is given. Find the total, the sport with maximum students, and the fraction who play it. Then choose a scale for a bar graph and give the height of the biggest bar. 5

Sport	Students
Kabaddi	35
Hockey	50
Tennis	50
Chess	60
Cricket	45

OR

Find the smallest number that leaves a remainder 1 when divided by 4, 5 and 6, and the next such number. 5

27. A pattern of squares: the first figure has 3 squares and each next figure has 5 more. Find the number of squares in the 12th figure, and the first figure that has more than 108 squares. 5

OR

A clock shows 5:00. Through how many degrees must the minute hand turn to make the time 5:20? Through how many degrees has the hour hand turned? Find the angle between the hands then. 5

END OF QUESTION PAPER Total: 50 Marks

MASTERMIND